

- 1. **Experiment:** Homologous Recombination and pCDH Library Construction
- 2. **Time:** 2025.12.17-2025.12.30
- 3. **Member:** Kaiqing Zhang, Yang Jin, Binxuan Zhang, Qiwen Jiang, Yue Fu, Zhangdong Liu
- 4. **Method:**

- (1) Library design and pCDH vector linearization
 - ① Nine prioritized *de novo* binder modules (A1-A3, B1-B3, and C1-C3) were organized into three fragment classes. The A, B, and C modules were combinatorially assembled to generate 27 A-B-C constructs for the first-round pCDH library. The pCDH-EF1-MCS-FcRTM+FcRECD-T2A-Puro-Kan (pCDH) vector was linearized by XbaI/BamHI digestion. The linearized vector served as the backbone for ordered assembly of the A, B, and C fragments using a commercial Vazyme homologous recombination system. The plasmid map showing the XbaI/BamHI restriction sites is shown below:

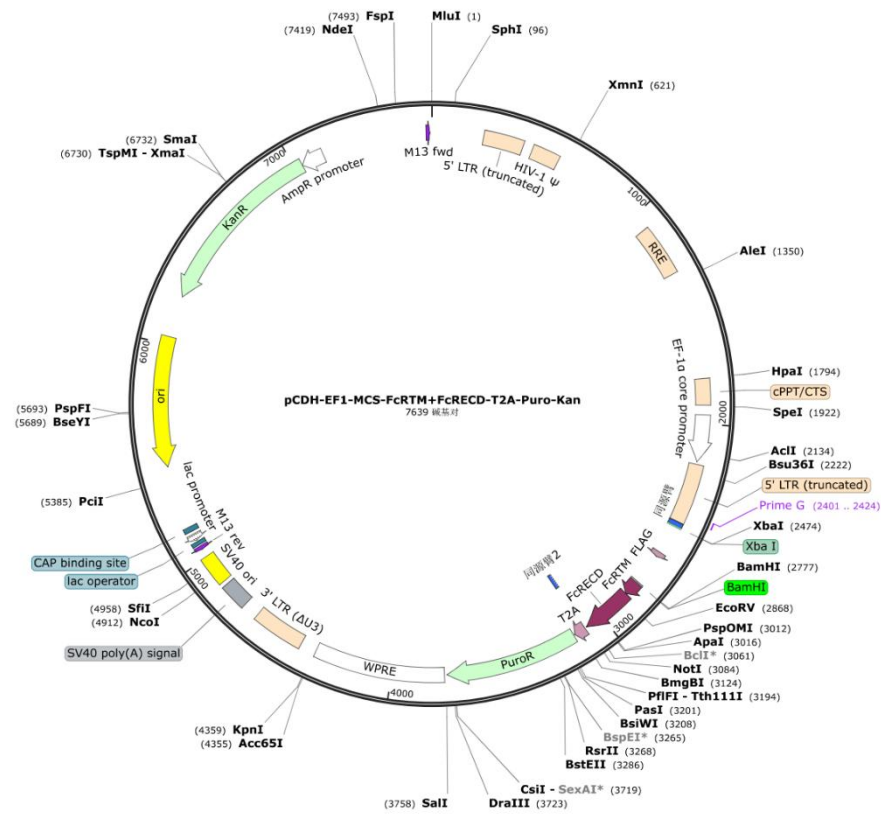


Figure 1. Plasmid map of the pCDH-EF1-MCS-FcRTM+FcRECD-T2A-Puro-Kan vector (pCDH).

- (2) Preparation of Linearized Vector
 - ① Suitable cloning sites are chosen for vector linearization. Sites lacking repetitive sequences and exhibiting GC content between 40% and 60% within the 20 bp regions upstream and downstream of the cloning site are preferentially selected.
 - ② The vector can be linearized via restriction enzyme digestion or inverse PCR amplification.
- (3) Acquisition of Insert Fragment
 - ① General primer design principles: Homologous sequences corresponding to the termini of the linearized vector are introduced at the 5'-ends of the forward and reverse amplification primers for the insert fragment. This ensures that the amplified insert fragment contains terminal homologous sequences at its 5' and 3' ends that match the corresponding sequences of the linearized vector.
 - ② Each insert fragment was designed with 30 bp terminal overlaps to enable ordered multi-fragment assembly. The 5' end of fragment A contained the sequence homologous to the left terminus of the linearized vector (AGCTGTGACCGGCGCCTACTCTAGA), followed by the signal peptide coding sequence

(ATGGCCCTCTGGATGAGGCTGCTGCCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCA). The 3' end of fragment A carried the Flag-tag coding sequence (GACTACAAGGACGACGACGACAAG), which also functioned as the overlap with fragment B.

- ③ The 5' end of fragment B carried the same Flag-tag sequence, whereas its 3' end contained the linker coding sequence (GGCGGGGGCAGCGCGGGGGCTCC) as the overlap with fragment C.
 - ④ The 5' end of fragment C carried the same linker sequence, and its 3' end contained the sequence homologous to the FcR transmembrane region of the vector (ATCATTGTGGCGGTGGTGAT).
- (4) Quantification of Linearized Vector and Insert Fragment
- ① Concentration determination: The concentrations of the gel-purified linearized vector and insert fragments were determined before recombination. The required DNA volumes were calculated from the measured concentrations as used in the original notebook, and the reaction was assembled to a final volume of 10 μ L.
 - ② Calculation of vector fragment usage: The optimal volumes of the linearized vector and insert fragments for the recombination reaction are calculated based on their respective concentrations measured after gel purification. The total reaction system is adjusted to 10 μ L with ddH₂O.
- (5) Recombination Reaction
- ① The required amounts of DNA for the recombination reaction are calculated based on the formulas provided above. To ensure pipetting accuracy, the linearized vector and insert fragments may be appropriately diluted prior to assembling the recombination reaction system. The volume of each component added should not be less than 1 μ L.
 - ② The following reaction system is prepared on ice:

Table 1. Homologous recombination reaction system.

Component	Volume (μ L)
Linearized vector (digestion product)	1.33
Insert A1/2/3 (PCR product)	0.43/0.47/0.47
Insert B1/2/3 (PCR product)	0.29/0.27/0.24
Insert C1/2/3 (PCR product)	0.29/0.36/0.37
2 $\times$ CE Mix	5
ddH ₂ O	to 10

- ③ The reaction mixture is gently mixed by pipetting (vortexing is avoided), followed by brief centrifugation to collect the liquid at the bottom of the tube.
 - ④ Preheat a water bath to 50 $^{\circ}$ C. The reaction is incubated at 50 $^{\circ}$ C for 30 min, then cooled to 4 $^{\circ}$ C or immediately placed on ice.
- (6) Transformation Procedure
- ① Cloning-competent *E. coli* Fast-T1 cells are thawed on ice.
 - ② The entire recombination product (10 μ L) is added to 100 μ L of competent cells. The mixture is gently mixed by flicking the tube (vortexing is avoided) and incubated on ice for 30 min.
 - ③ Cells are heat-shocked in a 42 $^{\circ}$ C water bath for 90 s, then immediately transferred to ice for 2 min.
 - ④ 900 μ L of LB medium is added, followed by incubation at 37 $^{\circ}$ C with shaking (220 rpm) for 1 h.
 - ⑤ After recovery, LB agar plates containing kanamycin (KLB plates) are pre-warmed in a 37 $^{\circ}$ C incubator. Cells are plated on kanamycin-containing LB agar and incubated at 37 $^{\circ}$ C for 12 h. Colonies were subsequently expanded in antibiotic-containing liquid LB medium, and plasmid DNA was extracted for downstream lentiviral packaging.
- (7) Verification of Recombinant Products
- ① Colony observation: After overnight incubation, hundreds of single colonies are observed on the recombinant product transformation plate.

- ② Colony PCR screening: Several colonies from the recombinant plate are selected for colony PCR. At least one universal sequencing primer from the vector is used for amplification. A band slightly larger than the insert fragment size is expected if the clone is correct.
- ③ Further validation: PCR-positive colonies are inoculated into liquid LB medium containing the appropriate antibiotic and cultured overnight. Plasmid DNA is extracted for restriction enzyme digestion analysis or directly subjected to Sanger sequencing.

5. Result:

- (1) The nine prioritized binder modules (A1-A3, B1-B3 and C1-C3) were combined into 27 A-B-C constructs, which were cloned into the XbaI/BamHI-linearized pCDH vector by multi-fragment homologous recombination. The 27 constructs were transformed into Fast-T1 competent cells and selected on kanamycin-containing LB agar; plasmid DNA prepared from the resulting colonies served as the source for subsequent lentiviral packaging (Table 3). Colony PCR and Sanger sequencing of the recovered clones confirmed the expected A-B-C assembly, and the full nucleotide sequence of each of the 27 constructs is listed in Table 2. Colony counts, colony PCR positive rates and the number of verified Sanger reads were not recorded, and no unrecorded values are stated here.

Table 2. Twenty-seven A-B-C constructs assembled into the pCDH vector by multi-fragment homologous recombination.

ID	Sequence
A1B1C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATGGAGG AAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCC AGGGAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCTCGAGGAGCTGCTG AAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAA GATTGCTGAGGTGGAGTTGGCATGGGAAAAGGGCGATTGGAAGGCCTATCTG GAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGTCCGCCGCA TTGCGCGCGAGAGGACTGGCCGCTCTCAGAGCAGAAGCCGAAAGGCTGGGC CTCGACCCTGCAGCAGTCAGGCTCGTGGCAGTGCTGCGGGTGGGAGACAGG GTGGTGGCTTTGGGAGAGCTGCCCCCAGAAGCATAACCCACGCTGACCCAG ACGATAGCAGAGTGCGCTTCGAGGCTGAGGTGCCCCGCTGGCGGGGGCAGCG GCGGGGGCTCCGCCCCGCCATACTCTGCCCCTGACCGTGCTGAAGGCGAAAC AGACGTATACGTGGATGGAAAGCTGGTGGCCACAGTGCCTGCTGGGGAGCA GGAGGTGACGGTGGAAGTGCCCCCGGAACCCATCGCGTGAGCCTCCGCAA TCGGAAGACTGGCGAGAGCAGAAGCTCCGATGTGGTCTTCCCTATCATTGTG GCGGTGGTGAT
A1B1C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATGGAGG AAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCC AGGGAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCTCGAGGAGCTGCTG AAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAA GATTGCTGAGGTGGAGTTGGCATGGGAAAAGGGCGATTGGAAGGCCTATCTG GAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGTCCGCCGCA TTGCGCGCGAGAGGACTGGCCGCTCTCAGAGCAGAAGCCGAAAGGCTGGGC CTCGACCCTGCAGCAGTCAGGCTCGTGGCAGTGCTGCGGGTGGGAGACAGG GTGGTGGCTTTGGGAGAGCTGCCCCCAGAAGCATAACCCACGCTGACCCAG ACGATAGCAGAGTGCGCTTCGAGGCTGAGGTGCCCCGCTGGCGGGGGCAGCG GCGGGGGCTCCGGAGCCGCCGCCAGACAAGCCGCAGCCCTGGACGCCGCCG

	CCGAAGCAGCCGAGGCTGCGGGAGACTTGGCCGCAGCTGCAGCAGCATATG CAGCCGCTGCAGCTCTCCTGGCAGCTCACGCCGCCACTCATCCGAATGGCGC CAACTTGGCCGCTTTGGCTGCTGAAGCCGCCGCTAGAGCCGCCGCCCTGGCT GCTCTGGCGGCCATCATTGTGGCGGTGGTGAT
A1B1C3	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATGGAGG AAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCC AGGGAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCTCGAGGAGCTGCTG AAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAA GATTGCTGAGGTGGAGTTGGCATGGGAAAAGGGCGATTGGAAGGCCTATCTG GAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGTCCGCCGCA TTGCGCGCGAGAGGACTGGCCGCTCTCAGAGCAGAAGCCGAAAGGCTGGGC CTCGACCCTGCAGCAGTCAGGCTCGTGGCAGTGCTGCGGGTGGGAGACAGG GTGGTGGCTTTGGGAGAGCTGCCCCCAGAAGCATAACCCACGCTGACCCAG ACGATAGCAGAGTGCGCTTCGAGGCTGAGGTGCCCGCTGGCGGGGGCAGCG GCGGGGGCTCCGGCTTTAGAGCAGCTGTCCCACTGAGAGTCGGGGACCCGG TCGAAATTAGGCTGGACGACGGGACAGTGGTTCAGGCAGGGTTCGATCGCGT GGAGGGCGACGTGGTCTCCGTGGCCCTCTTGAGCCACTCACCCGGGCGTG CACCGGGTGACCGTGAGGTTCTTGACCGAGCCTCCGCAGACCAGCGAGCCC AGACTGGTGGAGTTTCCCCCGGGTTCCCATCATTGTGGCGGTGGTGAT
A1B2C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATGGAGG AAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCC AGGGAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCTCGAGGAGCTGCTG AAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAA GATTGCTGAGGTGGAGTTGGCATGGGAAAAGGGCGATTGGAAGGCCTATCTG GAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGTCAGTAAAG GTCGTTACCTCTCAGTGCACCCTCCCAGATGGCACAGTGGTGACCGTAACTG TCTCTTGACCCGGACCCGGCGAGTACACGTGGGAGGTGAGCTTTTACGACGC CGACGATCCCAGCAATCGCACCCCTCCTTGACCCGTGACTTGACCTGTTCCC CCGACGACCCCGGCGGGGGCAGCGGCGGGGGCTCCGCCC GCCATACTCTGC CCCTGACCGTGCTGAAGGCGAAACAGACGTATACGTGGATGGAAAGCTGGT GGCCACAGTGCCTGCTGGGGAGCAGGAGGTGACGGTGGAAGTGCCCCCGG AACCCATCGCGTGAGCCTCCGCAATCGGAAGACTGGCGAGAGCAGAAGCTC CGATGTGGTCTTCCCTATCATTGTGGCGGTGGTGAT
A1B2C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATGGAGG AAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCC AGGGAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCTCGAGGAGCTGCTG AAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAA GATTGCTGAGGTGGAGTTGGCATGGGAAAAGGGCGATTGGAAGGCCTATCTG GAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGTCAGTAAAG GTCGTTACCTCTCAGTGCACCCTCCCAGATGGCACAGTGGTGACCGTAACTG TCTCTTGACCCGGACCCGGCGAGTACACGTGGGAGGTGAGCTTTTACGACGC CGACGATCCCAGCAATCGCACCCCTCCTTGACCCGTGACTTGACCTGTTCCC CCGACGACCCCGGCGGGGGCAGCGGCGGGGGCTCCGGAGCCGCCGCCAGA

	CAAGCCGCAGCCCTGGACGCCGCCGCCGAAGCAGCCGAGGCTGCGGGAGAC TTGGCCGCAGCTGCAGCAGCATATGCAGCCGCTGCAGCTCTCCTGGCAGCTC ACGCCGCCACTCATCCGAATGGCGCCAACCTTGGCCGCTTTGGCTGCTGAAGC CGCCGCTAGAGCCGCCGCCCTGGCTGCTCTGGCGGCCATCATTGTGGCGGTG GTGAT
A1B2C3	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATGGAGG AAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCC AGGGAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCTCGAGGAGCTGCTG AAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAA GATTGCTGAGGTGGAGTTGGCATGGGAAAAGGGCGATTGGAAGGCCTATCTG GAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGTCAGTAAAG GTCGTTACCTCTCAGTGCACCCCTCCAGATGGCACAGTGGTGACCGTAACTG TCTCTTGCACCGGACCCGGCGAGTACACGTGGGAGGTGAGCTTTTACGACGC CGACGATCCCAGCAATCGCACCCCTCCTTGACCCGTGACTTGCACCTGTTCCC CCGACGACCCCGGCGGGGGCAGCGGCGGGGGCTCCGGCTTTAGAGCAGCTG TCCCACTGAGAGTCGGGGACCCGGTGCAGAAATTAGGCTGGACGACGGGACAG TGGTTCCAGGCAGGGTCGATCGCGTGGAGGGCGACGTGGTCTCCGTGGCCCC TCTTGAGCCACTCACCCCGGGCGTGCACCGGGTGACCGTGAGGTTCTTGACC GAGCCTCCGCAGACCAGCGAGCCCAGACTGGTGGAGTTTCCCCCGGGTT CCCATCATTGTGGCGGTGGTGAT
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A1B3C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATGGAGG AAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCC AGGGAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCTCGAGGAGCTGCTG AAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAA GATTGCTGAGGTGGAGTTGGCATGGGAAAAGGGCGATTGGAAGGCCTATCTG GAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGGCACCATCA GCCGAACAGTTGGCCGCCCTTGCCGCAGAAGCCAGAGCTGCTTTGCTGGCCC AGGCAGCCGAACCTGGGCTGGACTCCGAGCGATCTCGAGTTGGAGGTGCGGC

	CAGTCCCCGAGGGAGTGCTGGCAATCGCACGCGCTACAGCCACCAGACCTG GCCAACCGCCCGTGAGATTGAGAGCCGAGGCCCTCCTGCGCCTGGGCGGGG GCAGCGGCGGGGGCTCCGGAGCCGCCAGACAAGCCGCAGCCCTGGACG CCGCCGCCGAAGCAGCCGAGGCTGCGGGAGACTTGGCCGCAGCTGCAGCAG CATATGCAGCCGCTGCAGCTCTCCTGGCAGCTCACGCCGCCACTCATCCGAAT GGCGCCAACTTGGCCGCTTTGGCTGCTGAAGCCGCCGCTAGAGCCGCCGCC TGGCTGCTCTGGCGGCCATCATTGTGGCGGTGGTGAT
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A2B1C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATTGACCC TGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAG AATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTGGCTAAGAAGAACC TTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTCG AGAAAGAGTTCGAGAAGGAGATTGAGAAGCTCAAAAAAGACTACAAGGAC GACGACGACAAGTCCGCCGCATTGCGCGCGAGAGGACTGGCCGCTCTCAGA GCAGAAGCCGAAAGGCTGGGCCTCGACCTGCAGCAGTCAGGCTCGTGCCA GTGCTGCGGGTGGGAGACAGGGTGGTGGCTTTGGGAGAGCTGCCCCCAGAA GCATACCCCCACGCTGACCCAGACGATAGCAGAGTGCGCTTCGAGGCTGAGG TGCCCGCTGGCGGGGGCAGCGCGGGGGCTCCGCCCCCATACTCTGCCCT GACCGTGCCTGAAGGCGAAACAGACGTATACGTGGATGGAAAGCTGGTGGC CACAGTGCCTGCTGGGGAGCAGGAGGTGACGGTGGAAGTGCCCCCGGAAC CCATCGCGTGAGCCTCCGCAATCGGAAGACTGGCGAGAGCAGAAGCTCCGA TGTGGTCTTCCCTATCATTGTGGCGGTGGTGAT
A2B1C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATTGACCC TGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAG AATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTGGCTAAGAAGAACC TTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTCG AGAAAGAGTTCGAGAAGGAGATTGAGAAGCTCAAAAAAGACTACAAGGAC GACGACGACAAGTCCGCCGCATTGCGCGCGAGAGGACTGGCCGCTCTCAGA

GCAGAAGCCGAAAGGCTGGGCCTCGACCCTGCAGCAGTCAGGCTCGTGGCA
GTGCTGCGGGTGGGAGACAGGGTGGTGGCTTTGGGAGAGCTGCCCCAGAA
GCATACCCCCACGCTGACCCAGACGATAGCAGAGTGCGCTTCGAGGCTGAGG
TGCCCGCTGGCGGGGGCAGCGGCGGGGGCTCCGGAGCCGCCGCCAGACAAG
CCGCAGCCCTGGACGCCGCCGCGAAGCAGCCGAGGCTGCGGGAGACTTG
CCGCAGCTGCAGCAGCATATGCAGCCGCTGCAGCTCTCCTGGCAGCTCACGC
CGCCACTCATCCGAATGGCGCCAACTTGGCCGCTTTGGCTGCTGAAGCCGCC
GCTAGAGCCGCCGCCCTGGCTGCTCTGGCGGCCATCATTGTGGCGGTGGTGA
T

A2B1C3 AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC
CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATTGACCC
TGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAG
AATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTGGCTAAGAAGAACC
TTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTTCG
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GACGACGACAAGTCCGCCGCATTGCGCGCGAGAGGACTGGCCGCTCTCAGA
GCAGAAGCCGAAAGGCTGGGCCTCGACCCTGCAGCAGTCAGGCTCGTGGCA
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GCATACCCCCACGCTGACCCAGACGATAGCAGAGTGCGCTTCGAGGCTGAGG
TGCCCGCTGGCGGGGGCAGCGGCGGGGGCTCCGGCTTTAGAGCAGCTGTCC
CACTGAGAGTCGGGGACCCGGTCGAAATTAGGCTGGACGACGGGACAGTGG
TTCCAGGCAGGGTCGATCGCGTGGAGGGCGACGTGGTCTCCGTGGCCCCTCT
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GCCTCCGCAGACCAGCGAGCCAGACTGGTGGAGTTCCCCCCCCGGGTTCCT
ATCATTGTGGCGGTGGTGAT

A2B2C1 AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC
CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATTGACCC
TGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAG
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TTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTTCG
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GACGACGACAAGTCAGTAAAGGTGTTACCTCTCAGTGCACCTCCAGATG
GCACAGTGGTGACCGTAACTGTCTCTTGACCCGGACCCGGCGAGTACACGTG
GGAGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGCACCCCTCCTTGC
ACCGTGACTTGACCTGTTCCCCCGACGACCCCGGCGGGGGCAGCGGCGGG
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AGACTGGCGAGAGCAGAAGCTCCGATGTGGTCTTCCCTATCATTGTGGCGGT
GGTGAT

A2B2C2 AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC
CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATTGACCC
TGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAG
AATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTGGCTAAGAAGAACC
TTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTTCG
AGAAAGAGTTTCGAGAAGGAGATTGAGAAGCTCAAAAAAGACTACAAGGAC

	GACGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGCACCCTCCCAGATG GCACAGTGGTGACCGTAACTGTCTCTTGACCCGGACCCGGCGAGTACACGTG GGAGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGCACCCCTCCTTGC ACCGTGACTTGACCTGTTCCCCCGACGACCCCGCGGGGGCAGCGGCGGG GGCTCCGGAGCCGCCGCCAGACAAGCCGCAGCCCTGGACGCCGCCGCCGAA GCAGCCGAGGCTGCGGGAGACTTGGCCGCAGCTGCAGCAGCATATGCAGCC GCTGCAGCTCTCCTGGCAGCTCACGCCGCCACTCATCCGAATGGCGCCAACT TGGCCGCTTTGGCTGCTGAAGCCGCCGCTAGAGCCGCCGCCCTGGCTGCTCT GGCGGCCATCATTGTGGCGGTGGTGAT
A2B2C3	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATTGACCC TGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAG AATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTGGCTAAGAAGAACC TTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTTCG AGAAAGAGTTTCGAGAAGGAGATTGAGAAGCTCAAAAAAGACTACAAGGAC GACGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGCACCCTCCCAGATG GCACAGTGGTGACCGTAACTGTCTCTTGACCCGGACCCGGCGAGTACACGTG GGAGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGCACCCCTCCTTGC ACCGTGACTTGACCTGTTCCCCCGACGACCCCGCGGGGGCAGCGGCGGG GGCTCCGGCTTTAGAGCAGCTGTCCCACTGAGAGTCGGGGACCCGGTCGAAA TTAGGCTGGACGACGGGACAGTGGTTCCAGGCAGGGTCGATCGCGTGAGAG GCGACGTGGTCTCCGTGGCCCTCTTGAGCCACTCACCCCGGGCGTGACCCG GGTGACCGTGAGGTTCTTGACCGAGCCTCCGCAGACCAGCGAGCCCAGACT GGTGAGTTTCCCCCCCCGGGTTCCCATCATTGTGGCGGTGGTGAT
A2B3C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATTGACCC TGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAG AATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTGGCTAAGAAGAACC TTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTTCG AGAAAGAGTTTCGAGAAGGAGATTGAGAAGCTCAAAAAAGACTACAAGGAC GACGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCGCAGAA GCCAGAGCTGCTTTGCTGGCCAGGCAGCCGAAGTGGGCTGGACTCCGAGC GATCTCGAGTTGGAGGTGCGGCCAGTCCCCGAGGGAGTGCTGGCAATCGCA CGCGCTACAGCCACCAGACCTGGCCAACCGCCCGTGAGATTGAGAGCCGAG GCCCTCCTGCGCCTGGGCGGGGGCAGCGGCGGGGGCTCCGCCCCCATACTC TGCCCCCTGACCGTGCCTGAAGGCGAAACAGACGTATACGTGGATGGAAAGCT GGTGCCACAGTGCCTGCTGGGGAGCAGGAGGTGACGGTGGAAGTGCCCCC CGGAACCCATCGCGTGAGCCTCCGCAATCGGAAGACTGGCGAGAGCAGAAG CTCCGATGTGGTCTTCCCTATCATTGTGGCGGTGGTGAT
A2B3C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATTGACCC TGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAG AATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTGGCTAAGAAGAACC TTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTTCG AGAAAGAGTTTCGAGAAGGAGATTGAGAAGCTCAAAAAAGACTACAAGGAC GACGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCGCAGAA

GCCAGAGCTGCTTTGCTGGCCCAGGCAGCCGAACTGGGCTGGACTCCGAGC
GATCTCGAGTTGGAGGTGCGGCCAGTCCCCGAGGGAGTGCTGGCAATCGCA
CGCGCTACAGCCACCAGACCTGGCCAACCGCCCGTGAGATTGAGAGCCGAG
GCCCTCCTGCGCCTGGGCGGGGGCAGCGGCGGGGGCTCCGGAGCCGCCGCC
AGACAAGCCGCAGCCCTGGACGCCGCCGCAAGCAGCCGAGGCTGCGGG
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GCTCACGCCGCCACTCATCCGAATGGCGCCAACTTGGCCGCTTTGGCTGCTG
AAGCCGCCGCTAGAGCCGCCGCCCTGGCTGCTCTGGCGGCCATCATTGTGGC
GGTGGTGAT

A2B3C3

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GACGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCGCAGAA
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CGCGCTACAGCCACCAGACCTGGCCAACCGCCCGTGAGATTGAGAGCCGAG
GCCCTCCTGCGCCTGGGCGGGGGCAGCGGCGGGGGCTCCGGCTTTAGAGCA
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GCCCCCTTTGAGCCACTCACCCCGGGCGTGCACCGGGTGACCGTGAGGTTCC
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A3B1C1

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CCGTGCCTGAAGGCGAAACAGACGTATACGTGGATGGAAAGCTGGTGGCCA
CAGTGCCTGCTGGGGAGCAGGAGGTGACGGTGGAAAGTGCCCCCGGAACCC
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A3B1C2

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GGCAAGGAAACTGTTTCATCGAGATCGCGGAAGAACTGGACTACAAGGACGA

A3B1C3

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GCTGCGGGTGGGAGACAGGGTGGTGGCTTTGGGAGAGCTGCCCCAGAAGC
ATACCCCCACGCTGACCCAGACGATAGCAGAGTGCGCTTCGAGGCTGAGGTG
CCCGCTGGCGGGGGCAGCGGCGGGGGCTCCGGAGCCGCCGCCAGACAAGCC
GCAGCCCTGGACGCCGCCGCCGAAGCAGCCGAGGCTGCGGGAGACTTGGCC
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CCACTCATCCGAATGGCGCCAACTTGGCCGCTTTGGCTGCTGAAGCCGCCG
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GCTGCGGGTGGGAGACAGGGTGGTGGCTTTGGGAGAGCTGCCCCAGAAGC
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CCAGGCAGGGTCGATCGCGTGGAGGGCGACGTGGTCTCCGTGGCCCCCTCTTG
AGCCACTCACCCCGGGCGTGCACCGGGTGACCGTGAGGTTCTTGACCGAGC
CTCCGCAGACCAGCGAGCCAGACTGGTGGAGTTTCCCCCGGGTTCCCAT
CATTGTGGCGGTGGTGAT

A3B2C1

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GCTATCAAACGTACGAGGAGGCATTGAAGAAGGACCCATCTAAGAAGCTGT
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GGCAAGGAAACTGTTTCATCGAGATCGCGGAAGAAGTGGACTACAAGGACGA
CGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGACCCCTCCCAGATGGC
ACAGTGGTGACCGTAACTGTCTCTTGACCGGACCCGGCGAGTACACGTGGG
AGGTGAGCTTTTACGACGCCGACGATCCAGCAATCGCACCCCTCCTTGAC
CGTGACTTGACCTGTTCCCCGACGACCCGGCGGGGGCAGCGGCGGGGG
CTCCGCCCCGCATACTCTGCCCCTGACCGTGCCTGAAGGCGAAACAGACGTA
TACGTGGATGGAAGCTGGTGGCCACAGTGCCTGCTGGGGAGCAGGAGGTG
ACGGTGGAAGTGCCCCCGGAACCCATCGCGTGAGCCTCCGCAATCGGAAG
ACTGGCGAGAGCAGAAGCTCCGATGTGGTCTTCCCTATCATTGTGGCGGTGG
TGAT

A3B2C2

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AAATCGAGAAGGAAGCCGAGAAAGCCTTCGAGGAGAAAAAATACGAAGAG
GCTATCAAACGTACGAGGAGGCATTGAAGAAGGACCCATCTAAGAAGCTGT
ACTACAAGGAAGGGATTGCCTTCGCATATTACGAACTGGGCAAAAAAGAGGA
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	CGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGCACCCTCCCAGATGGC ACAGTGGTGACCGTAACTGTCTCTTGACACCGGACCCGGCGAGTACACGTGGG AGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGCACCCCTCCTTGAC CGTGACTTGACCTGTTCCCCGACGACCCCGGCGGGGGCAGCGGCGGGGG CTCCGGAGCCGCCGAGACAAGCCGACGCCCTGGACGCCGCCGCCGAAGC AGCCGAGGCTGCGGGAGACTTGGCCGCAGCTGCAGCAGCATATGCAGCCGC TGCAGCTCTCCTGGCAGCTCACGCCGCCACTCATCCGAATGGCGCCAAGTTG GCCGCTTTGGCTGCTGAAGCCGCCGCTAGAGCCGCCGCCCTGGCTGCTCTGG CGGCCATCATTGTGGCGGTGGTGAT
A3B2C3	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAAGCAAAG AAATCGAGAAGGAAGCCGAGAAAGCCTTCGAGGAGAAAAAATACGAAGAG GCTATCAAAGTGTACGAGGAGGCATTGAAGAAGGACCCATCTAAGAAGCTGT ACTACAAGGAAGGGATTGCCTTCGCATATTACGAACTGGGCAAAAAAGAGGA GGCAAGGAAACTGTTTCATCGAGATCGCGGAAGAACTGGACTACAAGGACGA CGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGCACCCTCCCAGATGGC ACAGTGGTGACCGTAACTGTCTCTTGACACCGGACCCGGCGAGTACACGTGGG AGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGCACCCCTCCTTGAC CGTGACTTGACCTGTTCCCCGACGACCCCGGCGGGGGCAGCGGCGGGGG CTCCGGCTTTAGAGCAGCTGTCCCACTGAGAGTCGGGGACCCGGTCGAAATT AGGCTGGACGACGGGACAGTGGTTCCAGGCAGGGTCGATCGCGTGAGAGGC GACGTGGTCTCCGTGGCCCCCTCTTGAGCCACTCACCCCGGGCGTGACCGGG TGACCGTGAGGTTCTTGACCGAGCCTCCGCAGACCAGCGAGCCCAGACTGG TGGAGTTTCCCCCCCCGGGTTCCCATCATTGTGGCGGTGGTGAT
A3B3C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAAGCAAAG AAATCGAGAAGGAAGCCGAGAAAGCCTTCGAGGAGAAAAAATACGAAGAG GCTATCAAAGTGTACGAGGAGGCATTGAAGAAGGACCCATCTAAGAAGCTGT ACTACAAGGAAGGGATTGCCTTCGCATATTACGAACTGGGCAAAAAAGAGGA GGCAAGGAAACTGTTTCATCGAGATCGCGGAAGAACTGGACTACAAGGACGA CGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCGCAGAAGC CAGAGCTGCTTTGCTGGCCCAGGCAGCCGAAGTGGGCTGGACTCCGAGCGA TCTCGAGTTGGAGGTGCGGCCAGTCCCCGAGGGAGTGCTGGCAATCGCACG CGCTACAGCCACCAGACCTGGCCAACCGCCCGTGAGATTGAGAGCCGAGGC CCTCTGCGCCTGGGCGGGGGCAGCGGCGGGGGCTCCGCCGCCATACTCTG CCCCTGACCGTGCCTGAAGGCGAAACAGACGTATACGTGGATGGAAGCTG GTGGCCACAGTGCCTGCTGGGGAGCAGGAGGTGACGGTGGAAGTGCCCCC GGAACCCATCGCGTGAGCCTCCGCAATCGGAAGACTGGCGAGAGCAGAAGC TCCGATGTGGTCTTCCCTATCATTGTGGCGGTGGTGAT
A3B3C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC CTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAAGCAAAG AAATCGAGAAGGAAGCCGAGAAAGCCTTCGAGGAGAAAAAATACGAAGAG GCTATCAAAGTGTACGAGGAGGCATTGAAGAAGGACCCATCTAAGAAGCTGT ACTACAAGGAAGGGATTGCCTTCGCATATTACGAACTGGGCAAAAAAGAGGA GGCAAGGAAACTGTTTCATCGAGATCGCGGAAGAACTGGACTACAAGGACGA CGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCGCAGAAGC

CAGAGCTGCTTTGCTGGCCCAGGCAGCCGAAGTGGGCTGGACTCCGAGCGA
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 CGCTACAGCCACCAGACCTGGCCAACCGCCCGTGAGATTGAGAGCCGAGGC
 CCTCCTGCGCCTGGGCGGGGGCAGCGGCGGGGGCTCCGGAGCCGCCGCCAG
 ACAAGCCGCAGCCCTGGACGCCGCCGCGGAAGCAGCCGAGGCTGCGGGAG
 ACTTGGCCGCAGCTGCAGCAGCATATGCAGCCGCTGCAGCTCTCCTGGCAGC
 TCACGCCGCCACTCATCCGAATGGCGCCAAGTGGCCGCTTTGGCTGCTGAA
 GCCGCCGCTAGAGCCGCCGCCCTGGCTGCTCTGGCGGCCATCATTGTGGCGG
 TGGTGAT
 A3B3C3 AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCC
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 GCTATCAAAGTGTACGAGGAGGCATTGAAGAAGGACCCATCTAAGAAGCTGT
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 GGCAAGGAACTGTTTCATCGAGATCGCGGAAGAACTGGACTACAAGGACGA
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 CAGAGCTGCTTTGCTGGCCCAGGCAGCCGAAGTGGGCTGGACTCCGAGCGA
 TCTCGAGTTGGAGGTGCGGCCAGTCCCCGAGGGAGTGCTGGCAATCGCACG
 CGCTACAGCCACCAGACCTGGCCAACCGCCCGTGAGATTGAGAGCCGAGGC
 CCTCCTGCGCCTGGGCGGGGGCAGCGGCGGGGGCTCCGGCTTTAGAGCAGC
 TGTCCCACTGAGAGTCGGGGACCCGGTCGAAATTAGGCTGGACGACGGGAC
 AGTGGTTCCAGGCAGGGTCGATCGCGTGAGGGCGACGTGGTCTCCGTGGC
 CCTCTTGAGCCACTACCCCGGGCGTGCACCGGGTGACCGTGAGGTTCTG
 ACCGAGCCTCCGCAGACCAGCGAGCCCAGACTGGTGGAGTTTCCCCCCCCG
 GTTCCCATCATTGTGGCGGTGGTGAT

Table 3. Concentration of the 27 pCDH constructs after large-scale plasmid preparation.

Group	Concentration
TE	14.45
A1B1C1	462.31
A1B1C2	427.3
A1B1C3	506.18
A1B2C1	937.18
A1B2C2	588.62
A1B2C3	746.38
A1B3C1	728.55
A1B3C2	645.38
A2B1C1	526.07
A2B1C2	3979.52
A2B1C3	649.35
A2B2C1	869.11
A2B2C2	815.26
A2B2C3	5292.56
A2B3C1	1051.646667
A2B3C2	702.336667
A2B3C3	498.19

A3B1C1	1219.773333
A3B1C2	648.9133333
A3B1C3	935.6
A3B2C1	797.7333333
A3B2C2	836.53
A3B2C3	794.2333333
A3B3C1	1705.8
A3B3C2	1561.433333
A3B3C3	1246.486667
